

Problem 1:

Let X and Y be topological spaces, and let $\mathcal{C}(X, Y)$ have the compact-open topology. Prove that if Y is Hausdorff, then $\mathcal{C}(X, Y)$ is Hausdorff, and if Y is regular, then $\mathcal{C}(X, Y)$ is regular.

Claim

Let X, Y be topological spaces, and let $\mathcal{C}(X, Y)$ have the compact-open topology. Let U, V be open in Y .

Prove that if C is a compact set of X and $\bar{U} \subset V$, then the subbasis sets satisfy $\overline{S(C, U)} \subset S(C, V)$.

Proof.

Assume C is compact and $\bar{U} \subset V$. Let $f \in \overline{S(C, U)}$, and $x \in C$.

Assume for contradiction that $f(x) \notin \bar{U}$. Then consider $W = Y \setminus \bar{U}$ which is open neighborhood of $f(x)$.

Thus, consider $S(\{x\}, W)$. Since $\{x\}$ is finite it is compact thus $S(\{x\}, W)$ is an open neighborhood of f since $f(x) \in W$.

Thus, there exists a $g \in S(\{x\}, W)$ s.t. $g \in S(C, U)$. However, this is a contradiction since this implies that $g(x) \in W \cap U \subseteq W \cap \bar{U} = \emptyset$.

Therefore, $f(x) \in \bar{U} \subseteq V$. Thus, $f \in S(C, V)$. ■

Proof.

Assume Y is Hausdorff. Let $f, g \in \mathcal{C}(X, Y)$ s.t. $f \neq g$. Thus, there is an $x \in X$ s.t. $f(x) \neq g(x)$. Then, because Y is Hausdorff, you can find disjoint open neighborhoods U and V in Y for $f(x)$ and $g(x)$.

Now consider $\{x\}$ it is finite thus compact. Therefore, $S(\{x\}, U)$ is an open neighborhood of f and $S(\{x\}, V)$ is an open neighborhood of g .

Assume for contradiction that there is an $h \in S(\{x\}, U) \cap S(\{x\}, V)$. This implies that $h(x) \in U \cap V$ which is a contradiction. Thus, we have found disjoint neighborhoods of f, g . Thus, $\mathcal{C}(X, Y)$ is Hausdorff.

Assume Y is regular. Let $f \in \mathcal{C}(X, Y)$ and let U be an open neighborhood of f . Thus, there exists subbasis sets s.t.

$$f \in S(C_1, U_1) \cap S(C_2, U_2) \cap \cdots \cap S(C_n, U_n) \subseteq U$$

Let $y \in f(C_i)$, since Y is regular this implies there is an open neighborhood W_y s.t.

$y \in \overline{W_y} \subset U_i$.

Since $f(C_i)$ is compact and covered by these W_y 's. This implies there is a finite subcover s.t. $f(C_i) \subset W_{y_1} \cup \dots \cup W_{y_k}$.

Define $W_i = W_{y_1} \cup \dots \cup W_{y_k}$.

Since for all j $\overline{W_{y_j}} \subset U_i$. This implies that $\overline{W_i} \subset U_i$.

Now consider $S(C_i, W_i)$. Since $f(C_i) \subset W_i$ this implies that $f \in S(C_i, W_i)$.

Therefore,

$$f \in V = \bigcap_{i=1}^n S(C_i, W_i)$$

Now consider $\overline{V}$

$$\overline{V} = \overline{\bigcap_{i=1}^n S(C_i, W_i)} \subset \bigcap_{i=1}^n \overline{S(C_i, W_i)} \subset \bigcap_{i=1}^n S(C_i, U_i) \subset U$$

Therefore, but the claim above $f \in \overline{V} \subset U$. Thus, $\mathcal{C}(X, Y)$ is regular. ■

Problem 2:

Let A_n be a sequence of matrices in $GL(d, \mathbb{R})$, and let A be another matrix. Recall that each A_n and A can be viewed as linear maps $\mathbb{R}^d \rightarrow \mathbb{R}^d$, and consider each of the four following conditions:

- (i) The maps $A_n : \mathbb{R}^d \rightarrow \mathbb{R}^d$ converge *pointwise* to A .
- (ii) The maps $A_n : \mathbb{R}^d \rightarrow \mathbb{R}^d$ converge *uniformly* to A .
- (iii) The maps $A_n : \mathbb{R}^d \rightarrow \mathbb{R}^d$ converge *uniformly on compacts* to A .
- (iv) For every $1 \leq i, j \leq d$, the (i, j) entry of the matrix A_n converges to the (i, j) entry of A .

Three of these conditions are equivalent to each other, but the fourth is not equivalent to any of the other three. Which one is not the same?

Proof.

(i), (iii), (iv) are equivalent.

(i) $\implies$ (iv)

Assume (i). Let $e_1, \dots, e_d$ be the standard basis vectors. This implies that

$$A_n(e_j) \rightarrow A(e_j)$$

Consider $\pi_i : \mathbb{R}^d \rightarrow \mathbb{R}$. Since this is continuous in $\mathbb{R}^d$ this implies that

$$\pi_i(A_n(e_j)) = (A_n)_{ij} \rightarrow A_{ij}.$$

Therefore, (iv).

(iv) $\rightarrow$ (iii)

Assume (iv)

This implies that $|(A_n)_{ij} - A_{ij}| \rightarrow 0$.

Now consider the norm

$$\|A_n - A\| = \sqrt{\sum_{i,j} |(A_n)_{ij} - A_{ij}|^2}$$

Since there are finitely many terms for any ϵ we just pick the max N of all the terms thus $\|A_n - A\| \rightarrow 0$.

Now let K be a compact set in $\mathbb{R}^n$. This means it is bounded. Thus, there is an $M > 0$ s.t. $\|x\| < M$.

Therefore,

$$\|(A_n - A)(x)\| \leq \|A_n - A\| \|x\|$$

So, for $\sup K$ this implies that

$$\sup_{x \in K} \|(A_n - A)(x)\| \leq \|A_n - A\| \sum_{x \in K} \|x\| \leq M \|A_n - A\|$$

Since $\|A_n - A\| \rightarrow 0 \implies \sup_{x \in K} \|(A_n - A)(x)\| \rightarrow 0$.

Therefore, A_n converges uniformly on compacts to A .

(iii) $\rightarrow$ (i)

Assume (iii) compact convergence implies point wise convergence since $\{x\}$ is compact.

Why (ii) is not equivalent.

Consider $A_n = (1 + 1/n)I_d$. A_n converges point wise to the identity matrix. However, consider $\sup_{x \in \mathbb{R}^d} \|A_n(x) - I_d(x)\|$. Since $\mathbb{R}^d$ is unbounded consider a vector x s.t. $\|x\| = M$. Thus, $\|A_n(x) - I_d(x)\| = (1/n)M$, since we can pick an $M \in \mathbb{R}$. This implies that the supremum is infinite and does not exist. Thus, it does not uniformly converge. ■

Problem 3:

Let X be a compact Hausdorff space, and equip the set $\text{Homeo}(X)$ with the compact-open topology. Prove that $\text{Homeo}(X)$ is a topological group.

Proof.

Let $G = \text{Homeo}(X)$

Define $i : G \rightarrow G$ s.t. $i(x) = x^{-1}$.

Let C be compact and U open in X . Consider $W = S(C, U) \cap G$ which is open in G .

Consider $i^{-1}(W)$.

Consider $S(X \setminus U, X \setminus C) \cap G$. This is open since $X \setminus U$ is closed which implies compact, since X is compact Hausdorff, and $X \setminus C$ is open because C is compact in a Hausdorff space.

Let $f \in i^{-1}(W)$ then $f^{-1}(C) \subset U \implies X \setminus U \subset f^{-1}(X \setminus C) \implies f(X \setminus U) \subset X \setminus C \implies f \in S(X \setminus U, X \setminus C)$.

Let $f \in S(X \setminus U, X \setminus C) \implies f(X \setminus U) \subset X \setminus C \implies X \setminus U \subset f(X \setminus C) \implies f^{-1}(C) \subset U \implies f^{-1} \in W \implies f \in i^{-1}(W)$.

Thus, $i^{-1}(W) = S(X \setminus U, X \setminus C) \cap G$ which is open in G . Thus, i is continuous since we can build every other open set in G from this type.

Define $h : G \times G \rightarrow G$ s.t. $h(x, y) = x \circ y$.

Consider $h^{-1}(W)$. Let $(f, g) \in h^{-1}(W)$. Thus, $f \circ g \in W$.

Consider $g(C)$ which is compact and $f^{-1}(U)$ which is open.

Since $g(C) \subset f^{-1}(U)$. Since X is Hausdorff and compact thus locally compact. This implies that for $x \in g(C)$ there exists an open neighborhood W_x of x s.t. $\overline{W_x}$ is compact, and $\overline{W_x} \subset f^{-1}(U)$.

Thus, W_x form an open cover of $g(C)$ which is compact. Thus, there is a finite subcover $A = W_{x_1} \cup \dots \cup W_{x_n} \subset f^{-1}(U)$.

Since $\overline{W_{x_i}} \subset f^{-1}(U) \implies \overline{A} \subset f^{-1}(U)$.

Thus, consider $S(\overline{A}, U) \times S(C, A)$.

Let $(a, b) \in S(\overline{A}, U) \times S(C, A)$. Then we can see that $a \circ b(C) \subset a(A) \subset a(\overline{A}) \subset U$.

Thus, $a \circ b \in W$. Thus, h is continuous. ■

Problem 4:

Again, let X be a compact Hausdorff space, and let $\text{MCG}(X)$ denote the collection of connected components of $\text{Homeo}(X)$. One can define an operation $\cdot$ on $\text{MCG}(X)$ as follows: if C_1, C_2 are elements in $\text{MCG}(X)$, and $f_1 \in C_1, f_2 \in C_2$, then $C_1 \cdot C_2$ is the connected component of $\text{Homeo}(X)$ containing $f_1 \circ f_2$.

Prove that the operation $\cdot$ makes $\text{MCG}(X)$ into a group. *Note:* most of the work here is proving that the operation $\cdot$ is actually well-defined, i.e. proving that it does not matter which maps f_1, f_2 you pick in the definition above.

Proof.

Well-defined:

Consider C_1 and C_2 which are connected components of $\text{Homeo}(X)$. Since these are both connected this implies that $C_1 \times C_2$ is connected in $\text{Homeo}(X) \times \text{Homeo}(X)$.

Because the group operation map is continuous this implies that $h(C_1 \times C_2)$ is connected in $\text{Homeo}(X)$, so this is some subset of connected component let's call C_3 .

Thus, for $f, f' \in C_1$ and $g, g' \in C_2$, consider $h(f, g) = f \circ g \in C_3$ and $h(f', g') = f' \circ g' \in C_3$. Thus, the group operation is well defined.

Associativity:

This comes from function composition being associative.

Identity:

This is the connected component where the identity is because $[f] \cdot [id] = [f] = [id] \cdot [f]$.

Inverses:

For $f \in C_1$ the inverse is just which connected component f^{-1} is in. Thus, $[f] \cdot [f^{-1}] = [id] = [f^{-1}] \cdot [f]$. ■

Problem 5:

Let X be a completely regular space, and let $\beta(X)$ denote the Stone-Čech compactification of X . Prove that X is connected if and only if $\beta(X)$ is connected.

Proof.

Assume that $\beta(X)$ is disconnected.

Thus, $\beta(X) = A \cup B$ for disjoint open sets A, B . Thus, $A \cap X$ and $B \cap X$ are disjoint open sets in X . These are non-empty since X is dense in $\beta(X)$. Therefore, $X = A \cup B \cup X$. Thus, this implies that X is disconnected.

Assume that X is disconnected. Let $X = A \cup B$ for disjoint open sets of A, B .

Consider $f : X \rightarrow \{0, 1\}$ s.t. $f(x) = 1$ if $x \in A$ and 0 otherwise.

Let U be open in $\{0, 1\}$. So $U = \{1\}, \{0\}, \{1, 0\}$. In the first case $f^{-1}(\{1\}) = A, f^{-1}(\{0\}) = B, f^{-1}(\{0, 1\}) = X$. Thus, f is continuous.

Since $\{0, 1\}$ is compact and Hausdorff this implies that f can be extended to a continuous function $\hat{f} : \beta(X) \rightarrow \{0, 1\}$ s.t. $\hat{f}|_X = f$.

Also, $\hat{f}$ is still surjective since $\hat{f}|_X = f$ which is surjective. Thus, $\hat{f}^{-1}(0)$, and $\hat{f}^{-1}(1)$ are disjoint open sets that union to $\beta(X)$. Thus, $\beta(X)$ is disconnected. ■